

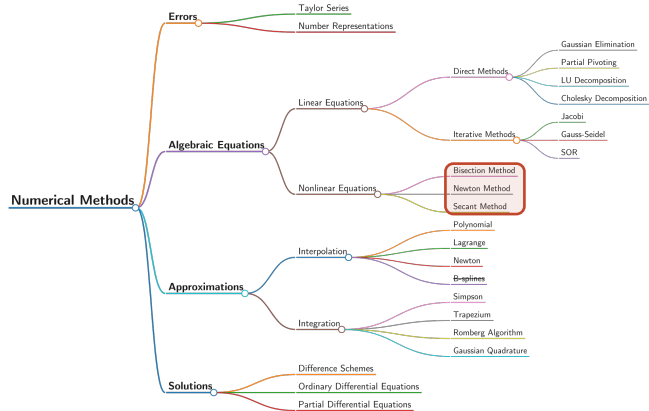
Solution Methods for Nonlinear Equations

CTMS-MAT-13 • Numerical Methods

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September 10, 2026

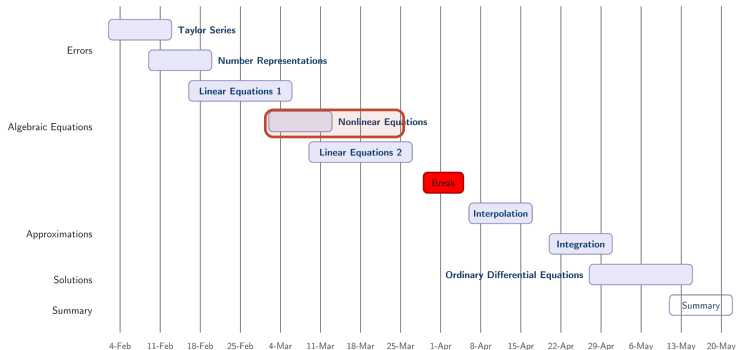
Where this sits in the course



Full outline: djps.github.io/docs/numericalmethods/intro/intro/

Schedule

Schedule 2026



Notes: djps.github.io/teaching/numerical_methods.pdf

Today

The problem

Bisection

Rates of convergence

Newton's method

Secant method

Systems

The problem

Why iterate at all

Some nonlinear equations factor:

$$f(x) = 12x^2 - 5x - 2 = (4x + 1)(3x - 2) = 0 \implies x = -\frac{1}{4}, \frac{2}{3}.$$

Most do not:

$$f(x) = \log(3 + x^4) + 356 x^{12} = 0.$$

TAKEAWAY

There is no formula to ask for. An iterative scheme finds *a* root near an initial guess —which root you get is part of what you choose.

Bisection

Bisection: trap the sign change

If $f \in C([a, b])$ and $f(a)f(b) < 0$, the Intermediate Value Theorem guarantees a root $r \in (a, b)$.

Split at $c = \frac{a+b}{2}$ and keep whichever half still changes sign. Repeat.

Theorem

After n steps the midpoint c_n satisfies

$$|r - c_n| < \frac{b - a}{2^n}.$$

Proof. $|r - c_1| < |a - c_1| = (b - a)/2$; the interval halves at each step, so $|r - c_n| < (b - a)/2^n$. $\square$

Bisection: how many steps

On $[a, b] = [0, 1]$, for an error below $2^{-20} \approx 10^{-6}$:

$$\frac{1}{2^n} < 2^{-20}$$
$$-\log_2(2^{-20}) < n \implies n \geq 20.$$

Twenty steps for six decimal places —and the count does not depend on f at all, only on the interval.

TAKEAWAY

Bisection is the only method here that cannot fail: guaranteed, predictable, and slow. One bit of accuracy per step.

Rates of convergence

Naming the speed

Let $x_n \rightarrow r$. Compare successive errors:

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - r|}{|x_n - r|} = \mu$$

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - r|}{|x_n - r|^q} < \mu$$

- $\mu \in (0, 1)$: **linear** —bisection has $\mu = \frac{1}{2}$;
- $\mu = 0$: **superlinear**;
- $\mu = 1$: **sublinear** (not very useful).

- order $q > 1$; $q = 2$ is **quadratic**, $q = 3$ **cubic**;
- q need not be an integer —the secant method has $q \approx 1.618$.

TAKEAWAY

Linear convergence adds a fixed number of digits per step; quadratic convergence *doubles* the digits.

Newton's method

Newton: follow the tangent

For $f \in C^1([a, b])$ and a guess x_0 , the Taylor approximation $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$ is the tangent line $t(x)$. Solve $t(x_1) = 0$ instead of $f(x_1) = 0$:

$$0 = f(x_0) + f'(x_0)(x_1 - x_0) \implies x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Iterating gives the **Newton–Raphson** scheme:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

requiring $f'(x_n) \neq 0$. When it converges, it converges to a genuine root.

Newton: convergence, global and local

Global

If $f(a)f(b) < 0$, $f' \neq 0$ on (a, b) , and f'' is continuous with constant sign, then f has exactly one root r and Newton *always* converges from

- $x_0 \in [a, r]$ if $f(a), f''$ share a sign;
- $x_0 \in [r, b]$ otherwise.

Local

If instead $f \in C^2([a, b])$ with $f(a)f(b) < 0$ and no critical points, then for x_0 *close enough* to r convergence is **quadratic**.

The convexity condition is what makes the global statement possible: it stops the tangent from overshooting to the wrong side.

Why it is quadratic

Expand about x_n and evaluate at the root, where $f(r) = 0$:

$$0 = f(x_n) + f'(x_n)(r - x_n) + \frac{1}{2}f''(\xi)(r - x_n)^2.$$

Divide by $f'(x_n)$ and regroup —the left side is exactly $r - x_{n+1}$:

$$r - x_{n+1} = -\frac{f''(\xi)}{2f'(x_n)}(r - x_n)^2 \quad \implies \quad |r - x_{n+1}| \leq \mu |r - x_n|^2,$$

with $\mu = \sup |f''(\xi)/2f'(x_n)|$.

□

Note where μ blows up: wherever f' is small.

Three ways Newton fails

1. **Run away** —the tangent at x_0 meets the axis far from the root, and the iterates diverge.
2. **Flat point** — $f'(x_n) \approx 0$ sends x_{n+1} off to infinity.
3. **Cycling** —the iterates alternate between two values, the convexity clause of the theorem having been violated.

TAKEAWAY

Quadratic convergence is a *local* promise. Bisection cannot fail but is slow; Newton is fast but needs a good start —in practice, bracket first, then polish with Newton.

Secant method

Secant: Newton without the derivative

If f is known only as data, or f' is expensive, freeze the derivative at some $m \leq n$ —the **modified Newton** method, $x_{n+1} = x_n - f(x_n)/f'(x_m)$.

Better: replace f' by a difference through the last two iterates,

$$f'(x_n) \approx \frac{f(x_{n-1}) - f(x_n)}{x_{n-1} - x_n},$$

which turns Newton's step into

$$x_{n+1} = x_n - f(x_n) \frac{x_{n-1} - x_n}{f(x_{n-1}) - f(x_n)}$$

Secant: the golden rate

Theorem

Let $f \in C^2([a, b])$ with $f(r) = 0$ and $f'(r) \neq 0$. Then

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - r|}{|x_n - r|^\tau} < M, \quad \tau = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

Superlinear but not quadratic —and each step costs one function evaluation instead of a function *and* a derivative.

TAKEAWAY

Per step the secant method is slower than Newton; per unit of work it is often faster.

Systems

Vector-valued equations

Now $\vec{f} : \mathbb{R}^m \rightarrow \mathbb{R}^n$, and we seek $\vec{f}(\vec{x}^*) = \vec{0}$. For $m = 3$, $n = 2$, $\vec{x} = (u, v, w)$:

$$\vec{f} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \begin{pmatrix} u^2 + vw \\ v + w^2/(w + u) \end{pmatrix}.$$

Newton needed f' ; the generalisation is the **Jacobian**, for $n = m$:

$$Df(\vec{x}) = \begin{pmatrix} \partial f_1 / \partial x_1 & \cdots & \partial f_1 / \partial x_n \\ \vdots & \ddots & \vdots \\ \partial f_n / \partial x_1 & \cdots & \partial f_n / \partial x_n \end{pmatrix}.$$

Newton for systems

$$\vec{x}_{n+1} = \vec{x}_n - [Df(\vec{x}_n)]^{-1} \vec{f}(\vec{x}_n).$$

The inverse is never formed. Instead solve one linear system per iteration:

$$Df(\vec{x}_n) \vec{y} = \vec{f}(\vec{x}_n), \quad \vec{y} = \vec{x}_n - \vec{x}_{n+1},$$

and update $\vec{x}_{n+1} = \vec{x}_n - \vec{y}$. The Jacobian must be non-singular.

TAKEAWAY

Nonlinear solving reduces to a sequence of linear solves—which is why the linear algebra lectures came first.

Bracketing is safe.
Tangents are fast.
Use both.

Summary

- Nonlinear equations are solved by iteration; the initial guess selects the root.
- Bisection needs only continuity and a sign change, and halves the error every step —linear, $\mu = \frac{1}{2}$, never fails.
- Newton follows the tangent: quadratic *locally*, but derailed by flat points, overshoot, or cycling.
- The secant method replaces f' with a difference quotient, converging at order $\tau = (1 + \sqrt{5})/2 \approx 1.618$.
- For systems, f' becomes the Jacobian and each step is a linear solve.

Take-home messages

1. **Bracket, then polish.** Bisect until the guess is safely local, then let Newton double the digits.
2. **Count work, not iterations.** A derivative-free step at order 1.618 can beat a quadratic step that costs twice as much.
3. **Watch f' .** Small derivatives are where the error constant $\mu = |f''/2f'|$ —and the method —blows up.
4. **Always cap the iteration count.** Divergence and cycling are normal outcomes, not exceptions; a stopping test on $|f(x_n)|$ alone is not enough.

Next: linear systems, direct and iterative.